

R6.65 – support 30, canonical source occurrences and Morita quotient

Append-only HT+NT continuity – formal EN report

Continuity lock and verification

R6.65 is the direct successor of R6.64, whose terminal bundle is locked by `SHA-256 218cb68e62e0...1df317f1`. The R6.65 principal verifier closes

`172078095/172078095` new checks, `492236793/492236793` cumulative checks.

The central class remains frozen:

`[ $\Omega_3$ ] = 1 STABLE`, `R7 CLOSED`.

Weight five: certified complete MITM at support 30

The canonical split $(i, j) \mid (k, l, m)$ covers the implicit universe $\binom{234}{5} = 5\,600\,390\,796$. The 232 disjoint slices $k = 0, \dots, 231$ cover their domains exactly once. The support-30 shell contains

`42916068`

unique canonical quintuples. Exhaustive semantic replay certifies

`171664504/171664504`

exact obligations. The topology pipeline is

`42916068  $\rightarrow$  2731144  $\rightarrow$  0  $\rightarrow$  0  $\rightarrow$  0`.

No candidate survives dual connectivity; no new H_3 sphere occurs at support 30. The retained 540-sphere carrier is unchanged.

Canonical source identities for all 540 occurrences

The former slot ordinals are removed. Occurrences are reconstructed directly from the frozen H3 carrier by their ordered source quintuple

$$\text{id}_{\text{src}}(S) = (\text{supp } S; i, j, k, l, m).$$

This reproduces exactly 246 support-16 and 294 support-17 occurrences, 594 overlap-four edges, 102 components, and

`$(-1, 0, +1) = (6, 42, 54)$` .

All 1620 occurrence flags are rewritten with these content identities; no ordinal `slot` field remains. This level is

`PROVEN-SOURCE-CONTENT-IDENTITIES`

relative to the frozen H3 generator carrier.

Exact no-go for a local raw-source K_h direction rule

R6.64 retains three transported direction kernels K_0, K_1, K_2 . R6.65 tests whether they can be reconstructed without the target chart π . On all 702 positive-representative pairs, the associated local two-component source bi-hypergraph is classified up to exact isomorphism. Only nine exact isomorphism types occur, and seven mix at least two labels among

$$\{\text{non-edge}, K_0, K_1, K_2\}.$$

Therefore every rule seeing only the local source neighbourhood of two components is insufficient:

$$\boxed{\text{local two-component raw-source } K_h = \text{REFUTED-TYPED}}.$$

This does not rule out global invariants, cocycles, higher flags or higher-arity constructions, which remain OPEN/NT. The retained K_{bf} channel nevertheless supplies a proven source-native anchor: the triples $(18, 33, 38)$ and $(69, 84, 89)$ are both lines in the same inherited d_0 direction family. Thus the semantic channel $(T9, T7, T5)$ remains anchored on that distinguished line without promoting the other two direction families.

Kernel groupoids, Burnside and Morita product

The retained source selector gives the component fibers, per block,

$$\boxed{1^{\times 10}, 2^{\times 11}, 3^{\times 5}, 4^{\times 1}},$$

and the occurrence fibers

$$\boxed{3^{\times 9}, 6, 9^{\times 6}, 12, 15^{\times 6}, 18, 21^{\times 3}}.$$

The “same fiber” relations therefore define fully source-native kernel equivalence groupoids.

Naive composition as spans of sets retains intermediate multiplicities 1, 2, 3, 4 and 3, 6, 9, 12, 15, 18, 21. Hence it does not provide a strict unit-multiplicity Burnside/Set-span action:

$$\boxed{\text{strict Set-span/Burnside} = \text{REFUTED-TYPED}}.$$

After quotienting by the kernel groupoids, however, every relevant intermediate fiber product is one orbit for the corresponding composition. The balanced Morita product of bibundles/profunctors exactly reproduces the retained transported K_h :

$$\boxed{\text{balanced kernel-groupoid Morita/profunctor} = \text{PROVEN}}.$$

A genuinely intrinsic Mackey structure remains OPEN/NT because source-native named direction/subgroup data are still absent.

Truth boundary and the 207 marker

The direct regular-action no-go, fibered groupoid and relational lift from R6.64 are retained. The marker

$$\boxed{207 = 200 + 7 = 9 \cdot 23}$$

remains SEPARATED/PENDING-MORPHISM: 200 and 7 are typed counts on different carriers. The full $D_{540} \times \{T5, T7, T9\}$ morphism remains OPEN/NT until all three raw-source direction labels and their global functoriality are explicit.

Safe R6.66 frontier

Retain $[\Omega_3]$ and all R6.65 carriers; extend weight five beyond support 30 only by certified complete MITM; retain the canonical 540 source occurrence identities and the source-native kernel quotient; retain the exact local two-component and strict Set-span/Burnside no-gos; seek a genuinely global or higher-arity source invariant, cocycle, flag or CW construction reconstructing the three direction families without π , using K_{bf} only as the proven source-native anchor for the $d_0/(T9, T7, T5)$ line; test whether the remaining two families are source-distinguishable or only a C_2 torsor; promote Mackey/Burnside only after source-native direction/subgroup data exist, while retaining the proven Morita/profunctor formulation for transported K_h ; keep 207 separated and keep R7 CLOSED until a source-exact relative higher cell acts nontrivially on $[\Omega_3]$.